TM65 mid-level 5 November 2026

LCA1-079V0321 - VFD Screw Air-cooled Chiller 78TR

Blue Star · Chillers · 274.3 kW Cooling

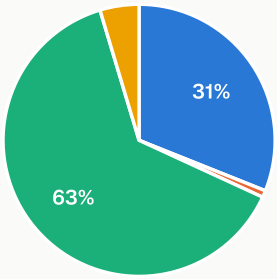
Total embodied carbon

106,053

$kgCO_2e$

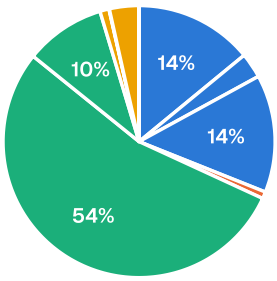
Refrigerant leakage during use (B1) is the largest contributor, at 53.9% of the total.

In tonnes	Per kg of product
106.05 tCO_2e	**34.42** $kgCO_2e/kg$
Per year of service	**Product weight**
4,242 $kgCO_2e/yr$ · 25 yrs	**3,081** kg · 97.3% covered



Product A1–A3	32,926	31.0%
Construction A4	867	0.8%
Use B1–B3	67,339	63.5%
End of life C1–C4	4,921	4.6%

Breakdown by life cycle stage



		kgCO₂e · share of total
A1 Material extraction	14,899	14.0%
A2 Transport	3,172	3.0%
A3 Manufacturing	14,855	14.0%
A4 Transport to site	867	0.8%
B1 Refrigerant leakage during use	57,200	53.9%
B3 Repair	10,139	9.6%
C1 Refrigerant leakage when decommissioning	1,144	1.1%
C2 Transport	53	<0.1%
C3 Waste processing	3,714	3.5%
C4 Disposal	11	<0.1%

Values include the TM65 buffer factor, except refrigerant leakage.

Calculation

A1	Material extraction	11,461.03
A2	Transport	2,440.15
A3	Manufacturing	11,426.58
A4	Transport to site	666.73
B1	Refrigerant leakage during use	57,200.00
B3	Repair	7,799.12
C1	Refrigerant leakage when decommissioning	1,144.00
C2	Transport	40.67
C3	Waste processing	2,856.65
C4	Disposal	8.23
	Subtotal	**36,699.15**
	Including buffer factor	**47,708.89**
B1 C1	Refrigerant leakage, in use and at end of life	58,344.00
Total, kgCO_2e		**106,052.89**

Product

Manufacturer	Blue Star
Type	Chillers
Capacity	274.3 kW Cooling
Weight	3,081.00 kg
Weight covered by materials	97.3%
Service life	25 years
Refrigerant	R134a, 80.00 kg
Factory	Wada, Mahārāshtra, India
Factory energy per unit	3,495.25 kWh

Materials

Sorted by embodied carbon

Material	Share of weight	A1, $kgCO_2e$	To factory
Electronic component	2.2%	3,374	8,246 km
Copper	20.0%	2,352	8,246 km
Steel, finished cold-rolled coil	24.6%	1,949	8,246 km
Steel, Section	19.6%	971	8,246 km
Steel, Engineering steel	17.1%	865	8,246 km
Aluminium sheet, Worldwide	1.5%	580	8,246 km
Steel (general or galvanised)	4.9%	447	8,246 km
Steel, Wire rod	4.0%	266	8,246 km
Aluminium cast, Worldwide	0.4%	158	8,246 km
Rubber	1.4%	126	8,246 km
Polyurethane foam	0.7%	96	8,246 km
Plastics (general)	0.9%	93	8,246 km
Cast iron	1.6%	74	8,246 km
Brass	0.3%	43	8,246 km
PVC	0.4%	42	8,246 km
Paint, Waterborne	0.2%	13	8,246 km
Silver	<0.1%	9	8,246 km
Ceramic	0.1%	2	8,246 km

Calculated with the CIBSE TM65 mid-level method. A self-assessed, deliberately cautious estimate for use when no Environmental Product Declaration is available.

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